

and main control boards, are built in-house.”

The company is headquartered in Kochi, Kerala, with manufacturing fully based in India at its Kalamassery factory.

“Currently, our setup is still in an early stage of scaling, using only around 3 to 4 GPUs for development work. Our near-term plan is to expand significantly, targeting the deployment of around 100 Nvidia Blackwell GPUs within our factory to support large-scale training and AI workloads. In parallel, we are also focusing on adding more advanced machining infrastructure, especially high-end CNC and cutting systems, to improve our in-house actuator manufacturing and overall production capability,” says Athil.

Testing and verification are central to the company’s workflow, with every stage evaluated through both simulation and real-world execution. The company uses software-based virtual training and simulation

systems to train and test robots with large amounts of data. At the same time, testing also takes place in its factory environment. The company has built an industrial training corridor, now expanded to around 24,000 square feet (approx. 2230 square metres), used for real-world robotic testing and development. In this setup, robots repeatedly learn tasks like picking and placing objects, correcting errors, and improving performance through repetition.

“In 2024, we raised around ₹45 million from US angel investors while continuing to operate on internally generated revenue from early projects and prototypes. Government funding has not come yet, but we are applying for research grants and collaborating with agencies like IDX,” explains Athil.

The company has signed multiple MoUs with major firms, including EY Global and SAP Germany, as well as several leading Indian IT and industrial companies.

It is also building academic partnerships, with an industrial humanoid R&D centre being set up in Bengaluru and joint development work starting next month. At the same time, the company is actively seeking channel partners across India to grow its presence in both industry and education.

Looking ahead, the company plans to build structured sales channels in the US and European markets as it expands operations. A US office is also planned before the end of 2026. Its long-term vision includes global expansion across North America, South America, and Europe, along with scaling production capacity. The company currently produces around 300 humanoid robots per month and aims to fully utilise this capacity within the next two years. Its vision is to scale indigenous humanoid robotics from India and supply advanced systems to global markets.

—Nidhi Agarwal

2 INDIA'S FIRST OPEN-ACCESS QUANTUM HARDWARE STARTUP BUILDING A 'COOPERATIVE FOR TECHNOLOGY'

India’s quantum ambitions are finally moving beyond theory, as a small team builds open-access hardware designed, tested, and manufactured locally.

A small team of quantum engineers and physicists, led by a former IBM India head, has built something India never had before. Qbit Force, founded in 2025, created open-access superconducting quantum computer testbeds with more than 50 per cent indigenous components, including the country’s first domestically designed dilution refrigerator. The model resembles the Amul cooperative approach applied to quantum technology. In the 1970s, Indian farmers came together to solve milk shortages by working collectively. Today, Qbit Force brings together

scientists, startups, students, defence labs, welders, goldsmiths, and forklift operators to build quantum hardware in an open, collaborative, and fully visible manner.

The reason for this startup is strategic and urgent. India has produced thousands of quantum-trained researchers through academic courses



India's first open-access superconducting quantum test bed in Amaravati